

SUMMARY OF THE CHANGES MADE

With screenshots or sample output of the program

Blessie Joy T. Estinopo
BSIT 1-2

PART 2 OF QUIZ 2

1. Added Main Navigation Menu

At the first two commits, I continuously improved my management menu which explains why the first two have different outputs. Until the last where I was able to create a cleaner and more organized main menu where users can choose to:

- View Products
- Search Product
- Filter by Category
- Manage Cart
- View Order History
- Exit the Program

```
=====
                        SHOPPING CART SYSTEM
=====
1. View Products
2. Search Product
3. Filter by Category
4. Cart Management
5. Order History
6. Exit
=====
Enter Choice: _
```

```
=====
                        PRODUCT LIST
=====
```

ID	Product Name	Category	Price	Stock
1.	Fuji Apple	Food	PHP 77.00	Stock: 130
2.	Gardenia Bread	Food	PHP 85.00	Stock: 45
3.	Fresh Milk	Food	PHP 105.00	Stock: 30
4.	Wireless Mouse	Electronics	PHP 500.00	Stock: 10
5.	Keyboard	Electronics	PHP 850.00	Stock: 4
6.	T-Shirt	Clothing	PHP 250.00	Stock: 20
7.	Jeans	Clothing	PHP 900.00	Stock: 8
8.	Bioderm soap	Household	PHP 42.00	Stock: 50
9.	Zondrox	Household	PHP 150.00	Stock: 30
10.	Keratin Plus Gold	Household	PHP 57.00	Stock: 5

```
=====
Enter Product ID (0 to Back): _
```

Added the choice of '0' so users can return to menu if clicked by mistake

2. Added Product Search Feature

I added a search function that allows users to enter a product name and display matching results. This helps users find products faster instead of manually checking the whole product list.

```
Enter Product Name to Search: bread
RESULT:
-----
 2. Gardenia Bread      Food      PHP    85.00  Stock:  45
Press any key to continue...
```

```
Enter Product Name to Search: d
RESULT:
-----
 2. Gardenia Bread      Food      PHP    85.00  Stock:  45
 5. Keyboard            Electronics PHP   850.00  Stock:   4
```

'No product found' appears when it's not in the menu or when users commit typo errors.

```
Enter Product Name to Search: egg
RESULT:
-----
No product found.
Press any key to continue...
```

3. Added Product Categories and Filtering

I used the Category field in the Product class and added category filtering options such as:

```

===== CATEGORY MENU =====
1. Food
2. Electronics
3. Clothing
4. Household
Enter Choice: 2

CATEGORY: Electronics
-----
4. Wireless Mouse      Electronics  PHP    500.00  Stock:  10
5. Keyboard           Electronics  PHP    850.00  Stock:   4

Press any key to continue...

```

Users can now display products by category.

4. Added Cart Management Menu

I improved the cart system by allowing users to manage their cart before checkout. The cart menu now includes:

```

===== CART MANAGEMENT =====
1. View Cart
2. Update Quantity
3. Remove Item
4. Clear Cart
5. Checkout
6. Back
Enter Choice: _

```

5. Improved Stock Management

I updated the stock handling so that:

- Stocks decrease when items are added to cart
- Stocks return when items are removed or cart is cleared
- Stocks adjust properly when cart quantity is updated

```

Console.Write("Enter New Quantity: ");

int newQty;

if (!int.TryParse(Console.ReadLine(), out newQty) || newQty < 0)
{
    Invalid();
    return;
}

int oldQty = cart[id - 1];

if (newQty > oldQty)
{
    int add = newQty - oldQty;

    if (add > products[id - 1].RemainingStock)
    {
        Console.WriteLine("Not enough stock.");
        Pause();
        return;
    }

    products[id - 1].RemainingStock -= add;
}
else
{
    products[id - 1].RemainingStock += (oldQty - newQty);
}

cart[id - 1] = newQty;

```

6. Added Checkout Payment Validation

I added payment validation during checkout. The system now checks that:

- Payment input is numeric
- Payment is enough to cover the final total

If payment is invalid or insufficient, the system asks again until correct payment is entered.

```

Final Total:                               PHP 525.00
-----
Enter payment amount: PHP 500
Insufficient / Invalid payment.
Enter payment amount: PHP 

```

7. Added Receipt Number and Date/Time

I improved the receipt by displaying:

- Receipt Number
- Current Date and Time
- Purchased Items
- Grand Total
- Discount
- Final Total

- Payment
- Change

```

=====
                        RECEIPT
Receipt No: 0002
Date: May 01, 2026 09:53 am
=====
Food      Fresh Milk      15   PHP  1575.00
Electronics Wireless Mouse   5   PHP  2500.00
Electronics Keyboard       3   PHP  2550.00
Clothing   Jeans           8   PHP  7200.00
-----
Grand Total:                      PHP 13825.00
Discount (10%):                  -PHP 1382.50
-----
Final Total:                      PHP 12442.50
-----
Enter payment amount: PHP 12500
Payment:                        PHP 12500.00
Change:                         PHP 57.50
=====
Thank you for your purchase!

```

8. Added Discount Logic

I added an automatic 10% discount when the grand total reaches PHP 5000 or above.

```

-----
Grand Total:                      PHP 13825.00
Discount (10%):                  -PHP 1382.50
-----
Final Total:                      PHP 12442.50
-----

```

9. Added Low Stock Alert

After checkout, the system checks all products and displays items with stock less than or equal to 5 so restocking can be monitored.

```

=====
Thank you for your purchase!

LOW STOCK ALERT
-----
Wireless Mouse has only 5 stock(s) left.
Keyboard has only 0 stock(s) left.
Jeans has only 0 stock(s) left.
Shampoo has only 5 stock(s) left.

Press any key to continue...

```

10. Added Order History

I created an order history array that stores completed transactions during the program run. Users can view previous receipts and totals anytime through the menu.

```
===== ORDER HISTORY =====  
Receipt #0001 - PHP 1060.00  
Receipt #0002 - PHP 12442.50  
  
Press any key to continue...
```

11. Improved Input Validation

I used `int.TryParse()` and `double.TryParse()` to prevent crashes caused by invalid input. I also improved Y/N prompts so they only accept valid responses.

```
if (!int.TryParse(Console.ReadLine(), out choice))  
{  
    Invalid();  
    return;  
}
```

```
if (double.TryParse(Console.ReadLine(), out payment))  
{  
    if (payment >= finalTotal)  
        break;  
}
```

```
static bool AskYesNo(string message)  
{  
    while (true)  
    {  
        Console.Write(message);  
        string input = Console.ReadLine().Trim().ToUpper();  
  
        if (input == "Y")  
            return true;  
  
        if (input == "N")  
            return false;  
  
        Console.WriteLine("Please enter Y or N only.");  
    }  
}
```

12. Improved Console Layout

I reformatted menus, product tables, receipts, and outputs to make the interface cleaner, more readable, and user-friendly.